



ESCUELA DE INGENIERÍA
FACULTAD DE INGENIERÍA

EDUCACIÓN
PROFESIONAL

Scene Text Recognition

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Today's schedule

01

What is Scene Text Recognition?

02

Applications

03

Stages

04

Datasets

05

Evaluation

06

ABCNet

What is Scene Text Recognition (STR)?

Recognition (detection and reading) of text in **natural scenes**, as opposed to, Optical Character Recognition (OCR) in scanned documents.





Applications

- Autonomous driving
- Multimedia retrieval
- Digitization of handwritten documents
- Augmented Reality
- Visually impaired assistance

STR vs OCR in scanned documents

OCR in scanned documents

Scene Text Recognition (STR)

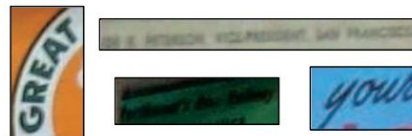
Full image

ITEM/DESC.	QTY	U.PRICE	RM(TOTAL)
4132	1	20.00	20.00
CROCS 300X17 TUBES			
SUB TOTAL :			20.00
ROUNDING ADJUSTMENT :			0.00
GRAND TOTAL :			20.00
CASH :			20.00
CHANGE :			0.00



Text instance

0.00 CROCS 300X17 TUBES INVOICE
CHANGE ROUNDING ADJUSTMENT 20.00



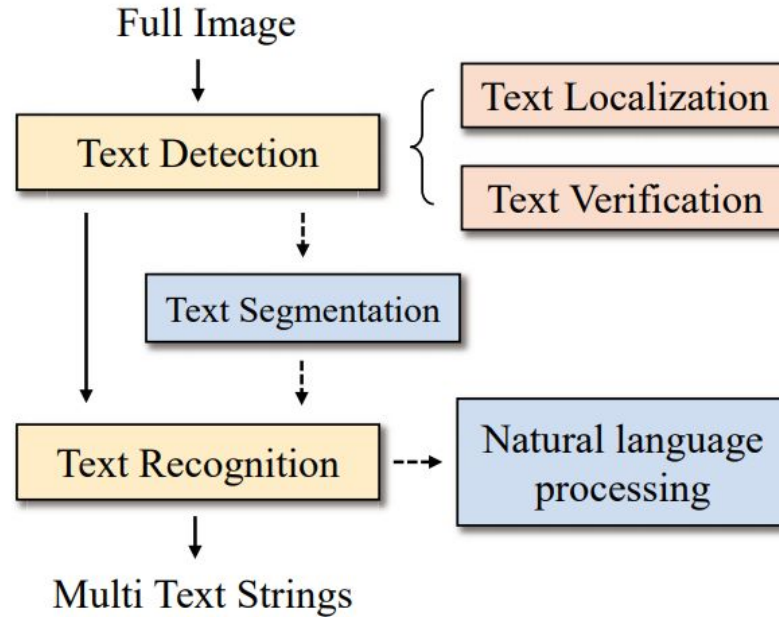


STR vs OCR in scanned documents

Challenges

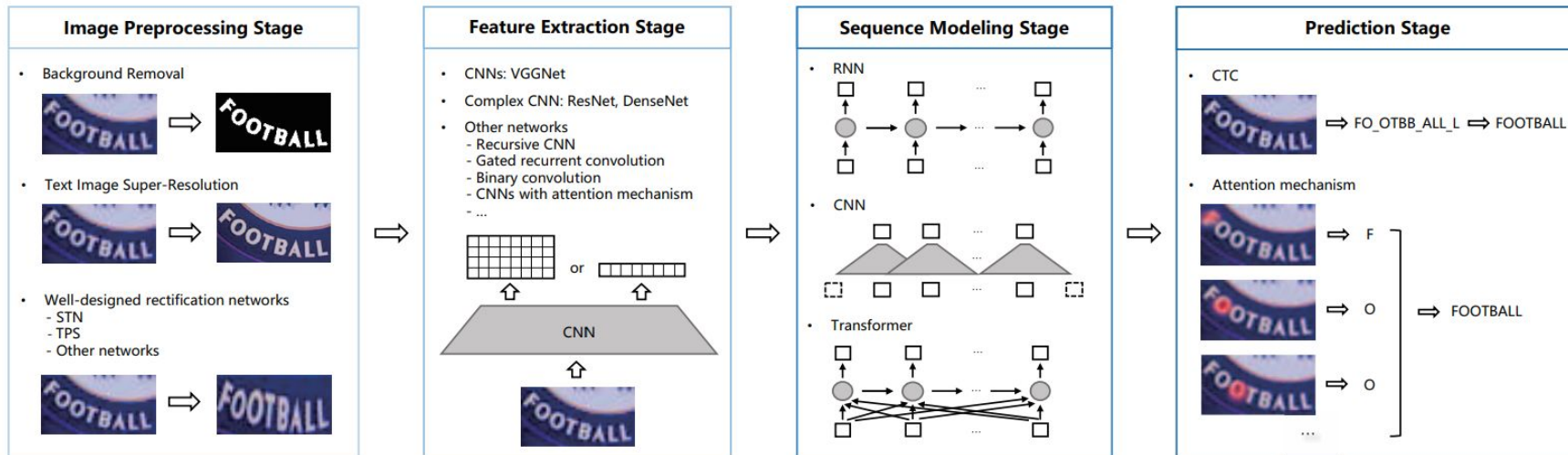
- Background
 - Text can appear on anything
 - Texture of background can be similar to text
- Form
 - Multiple colors
 - Irregular fonts
 - Different sizes
 - Diverse orientations
- Noise
 - Nonuniform illumination
 - Low resolution
 - Motion blurring

Stages



Stages

Text Recognition





Datasets

- Origin
 - Realistic
 - Synthetic
- Form of text
 - Regular
 - Irregular
- Script
 - Latin
 - Chinese
 - ...
 - Multilingual

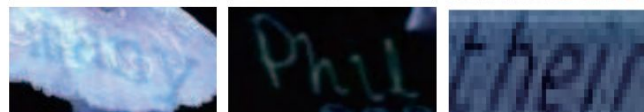
Datasets

Synthetic datasets

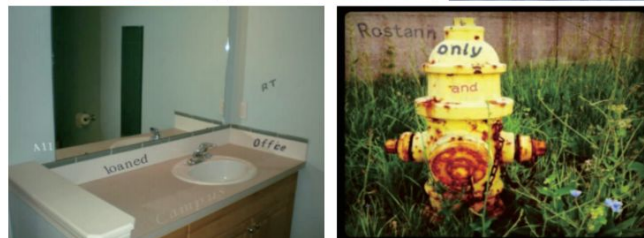
Synth90k



Synth Text



Verisimilar
Synthesis



UnrealText



Datasets

Realistic regular latin datasets



Datasets

Realistic irregular latin datasets

SVT-P



CUTE80



IC15



COCO-Text



Total-Text





Datasets Summary

Datasets	Language	Images			Instances			Lexicon	Char-Level Label	Type
		Total	Train	Test	Total	Train	Test			
Synth90k [221]	English	~9000000	-	-	~9000000	-	-	✗	✗	Regular
SynthText [217]	English	~6000000	-	-	~6000000	-	-	✗	✓	Regular
Verisimilar Synthesis [143]	English	-	-	-	~5000000	-	-	✗	✗	Regular
UnrealText [222]	English	~600000	-	-	~12000000	-	-	✗	✓	Regular
IIIT5K [69]	English	1120	380	740	5000	2000	3000	50 and 1k	✓	Regular
SVT [68], [180]	English	350	100	250	725	211	514	50	✗	Regular
IC03 [6]	English	509	258	251	2268	1157	1111	50, 1k and full	✓	Regular
IC11 [8]	English	522	420	102	4501	3583	918	✗	✓	Regular
IC13 [9]	English	561	420	141	5003	3564	1439	✗	✓	Regular
SVHN [223]	Digits	600000	573968	26032	600000	573968	26032	✗	✓	Regular
SVT-P [124]	English	238	0	238	639	0	639	50 and full	✗	Irregular
CUTE80 [224]	English	80	0	80	288	0	288	✗	✗	Irregular
IC15 [10]	English	1500	1000	500	6545	4468	2077	✗	✗	Irregular
COCO-Text [225]	English	63686	43686	10000	145859	118309	27550	✗	✗	Irregular
Total-Text [226]	English	1555	1255	300	11459	11166	293	✗	✗	Irregular
RCTW-17 [227]	Chinese/English	12514	11514	1000	-	-	-	✗	✗	Regular
MTWI [228]	Chinese/English	20000	10000	10000	290206	141476	148730	✗	✗	Regular
CTW [229]	Chinese/English	32285	25887	3269	1018402	812872	103519	✗	✓	Regular
SCUT-CTW1500 [230]	Chinese/English	1500	1000	500	10751	7683	3068	✗	✗	Irregular
LSVT [12], [231]	Chinese/English	450000	30000	20000	-	-	-	✗	✗	Irregular
ArI [232]	Chinese/English	10166	5603	4563	98455	50029	48426	✗	✗	Irregular
ReCTS-25k [232]	Chinese/English	25000	20000	5000	119713	108924	10789	✗	✓	Irregular
MLT [233]	Multilingual	20000	10000	10000	191639	89177	102462	✗	✗	Irregular

Evaluation

Text Detection

- Precision
- Recall
- F-score or Hmean

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$Hmean = \frac{2 * Precision * Recall}{Precision + Recall}$$

TP, FP and FN are computed based on the IoU of predicted bounding boxes and ground truth

Evaluation

Text Recognition

- Word Recognition Accuracy (WRA)

$$WRA = \frac{W_r}{W},$$

W Total number of words

W_r Correctly recognized words

- Normalized Edit Distance (NED)

$$NED = \frac{1}{N} \sum_{i=1}^N D(s_i, \hat{s}_i) / \max(l_i, \hat{l}_i)$$

$D(.)$ levenshtein distance

s_i predicted text

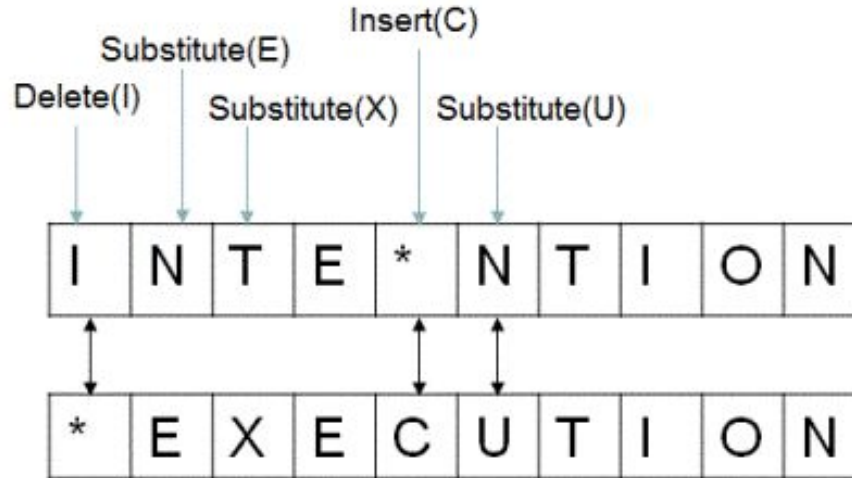
$\hat{s}_i$ ground truth

N total number of text lines

l_i length of predicted text

$\hat{l}_i$ length of ground truth

Evaluation Levenshtein



The Levenshtein distance between **intention** and **execution** is 5 because that is the minimum amount of operations needed to go from one string to the other

Evaluation

End-to-End

- Precision
- Recall
- F-score or Hmean

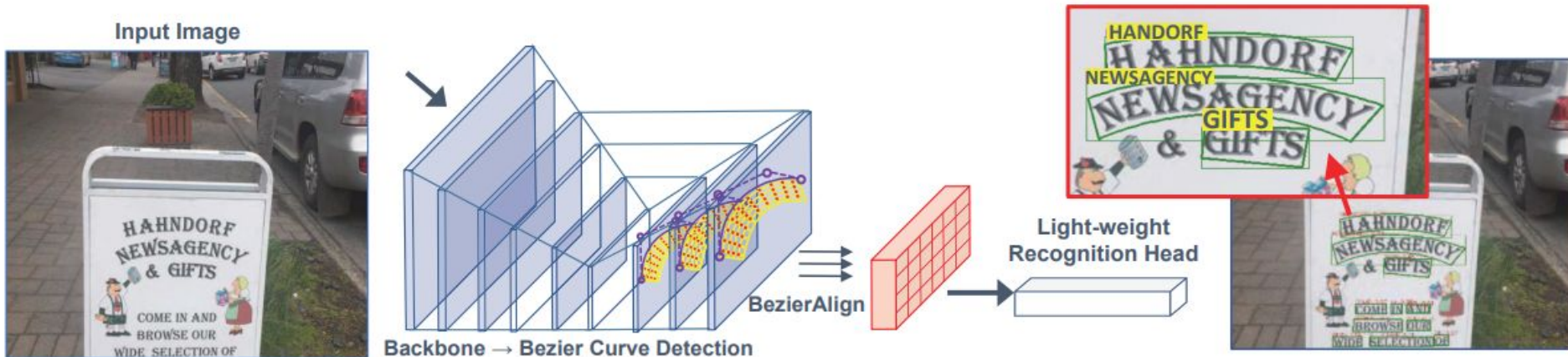
$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$Hmean = \frac{2 * Precision * Recall}{Precision + Recall}$$

TP, FP and FN are computed based on the IoU of predicted bounding boxes and ground truth and the value of the predicted text and the ground truth text

ABCNet: Adaptive Bezier Curve



ABCNet Comparison

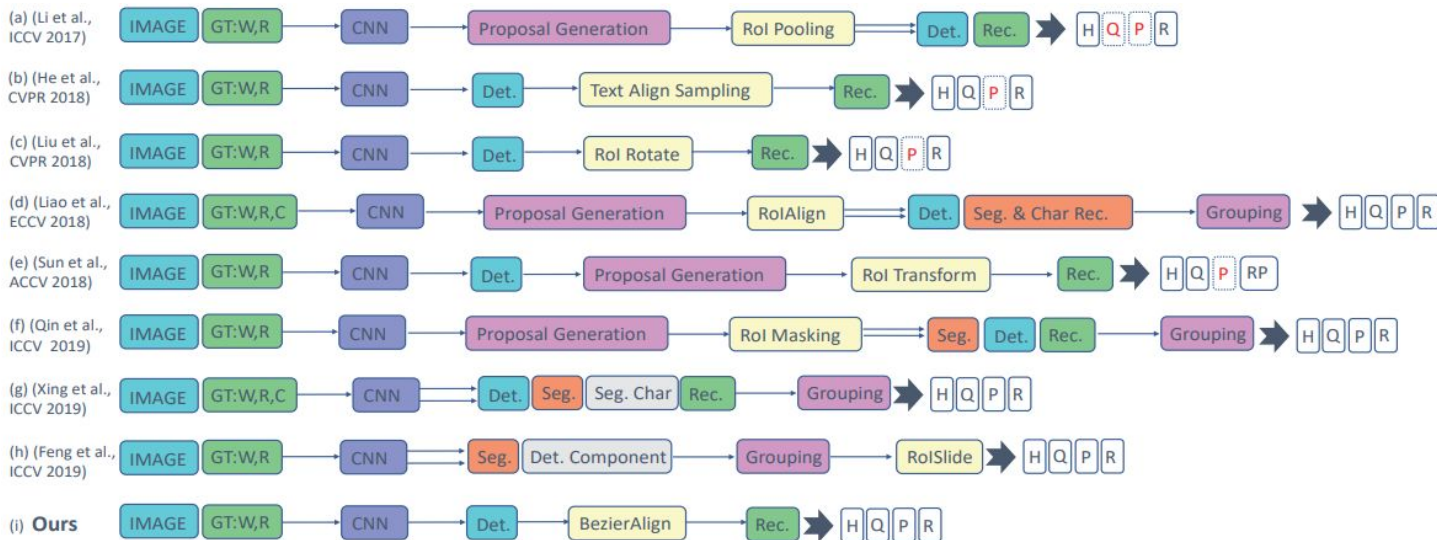
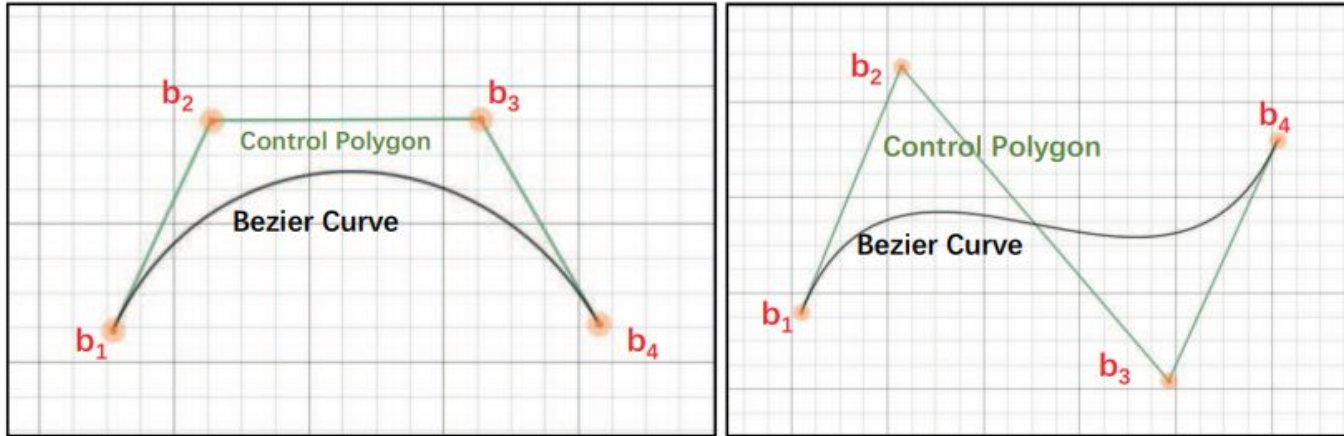


Figure 2. Overview of some end-to-end scene text spotting methods that are most relevant to ours. Inside the GT (ground-truth) box, ‘W’, ‘R’, and ‘C’ represent word-level annotation, text content, and character-level annotation, respectively. ‘H’, ‘Q’, and ‘P’ represent that the method is able to detect horizontal, quadrilateral, and arbitrarily-shaped text, respectively. ‘RP’ means that the method can recognize the curved text inside a quadrilateral box. ‘R’: recognition; ‘BBox’: bounding box. Dashed box represents the shape of the text which the method is unable to detect.

ABCNet

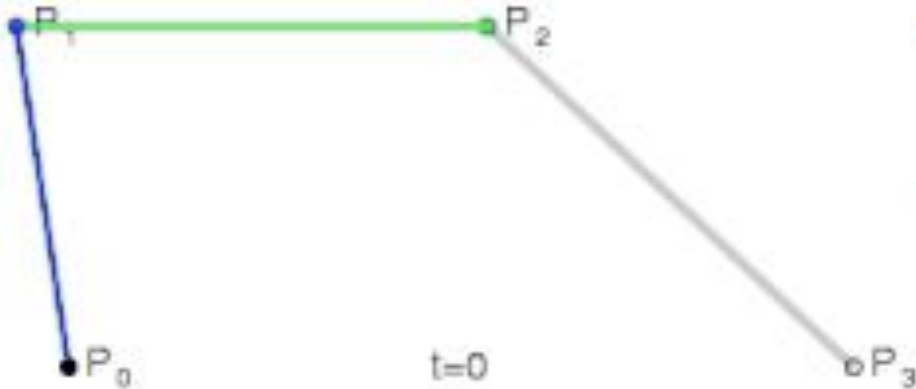
Cubic bezier curve



4 control points are enough to describe most of irregular scene texts

ABCNet

Bezier Curve



$$c(t) = \sum_{i=0}^n b_i B_{i,n}(t), 0 \leq t \leq 1$$

$$B_{i,n}(t) = \binom{n}{i} t^i (1-t)^{n-i}, i = 0, \dots, n$$

Using $c(t)$ we can obtain the coordinates of any point in the curve for t between 0 and 1

ABCNet

Polygon annotation vs Bezier



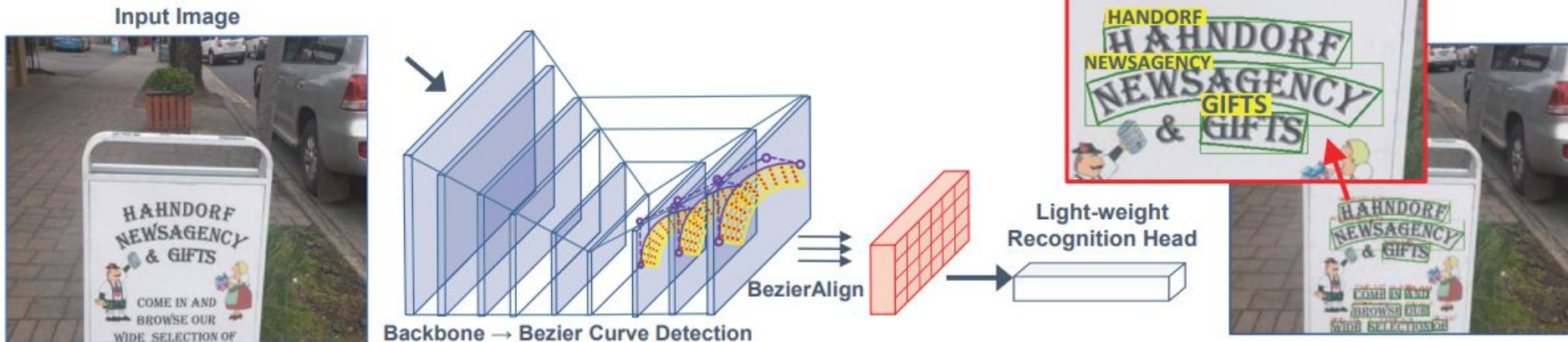
ABCNet

Bezier annotation



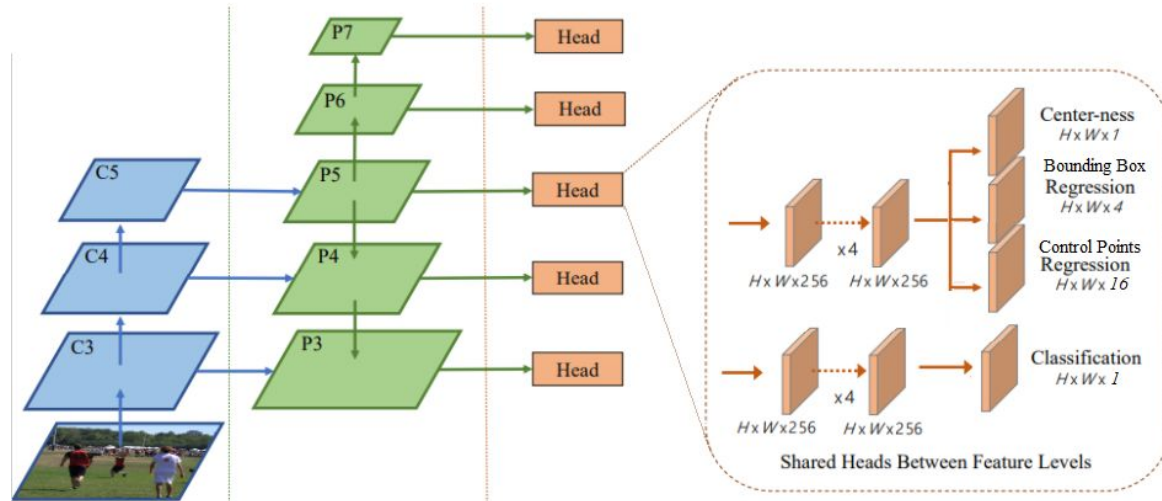
ABCNet

Full pipeline



ABCNet

Backbone+FPN+RPN (FCOS based)



ABCNet

Regression Losses

Bounding Box



IOU Loss

Center-ness



Binary Cross Entropy

Control Points



Smooth L1



ABCNet

Before pooling

- Application of NMS to proposals
 - Far from center proposals are usually of lower quality than close to center ones
 - Score of proposal is multiplied by the predicted center-ness. This greatly improves the results

ABCNet

Bezier Align



(a) Horizontal sampling.



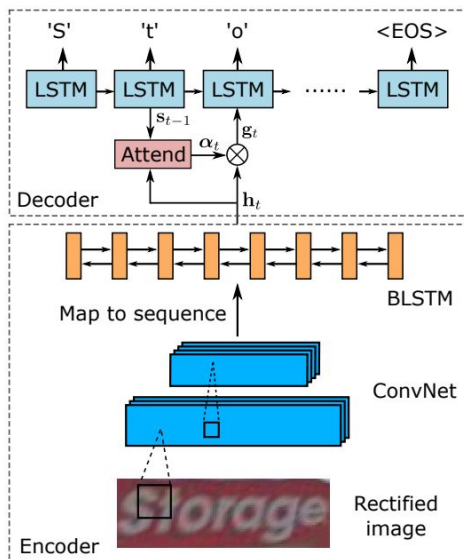
(b) Quadrilateral sampling.



(c) BezierAlign.

ABCNet

Attention based recognizer



$$e_{t,i} = \mathbf{w}^T \tanh(\mathbf{W}\mathbf{s}_{t-1} + \mathbf{V}\mathbf{h}_i + b)$$

$$\alpha_{t,i} = \exp(e_{t,i}) / \sum_{i'=1}^n \exp(e_{t,i'})$$

$$\mathbf{g}_t = \sum_{i=1}^n \alpha_{t,i} \mathbf{h}_i$$

$$(\mathbf{x}_t, \mathbf{s}_t) = \text{rnn}(\mathbf{s}_{t-1}, (\mathbf{g}_t, f(y_{t-1})))$$

$$p(y_t) = \text{softmax}(\mathbf{W}_o \mathbf{x}_t + b_o)$$

$$y_t \sim p(y_t)$$



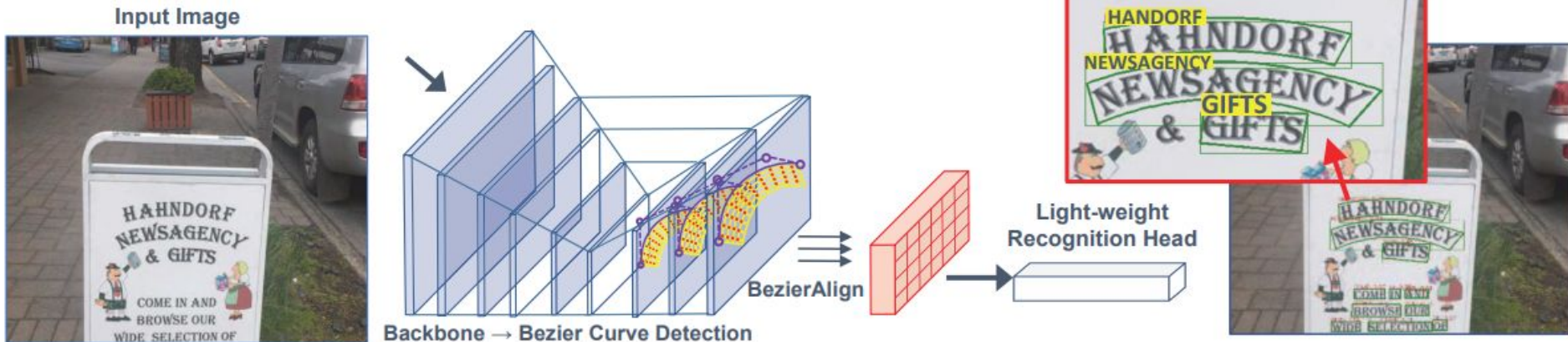
ABCNet

Recognition Loss

- Cross-entropy
 - Each symbol in our alphabet corresponds to a class
 - There are two extra classes
 - Unseen character
 - End of Sequence <EOS>

ABCNet

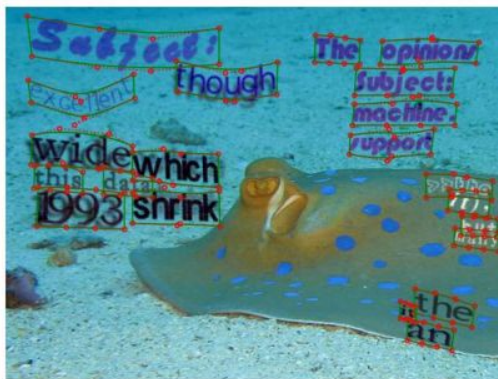
Full pipeline



ABCNet

Training datasets

- 15k images from COCO-Text
- 7k images from ICDAR-MLT
- 150k synthesized images



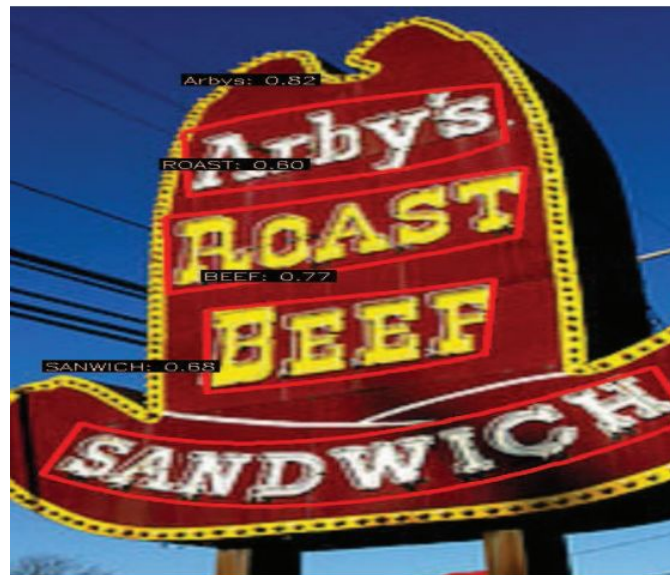
ABCNet

Quantitative results on Total-Text

Method	Data	Backbone	F-measure		FPS
			None	Full	
TextBoxes [21]	SynText800k, IC13, IC15, TT	ResNet-50-FPN	36.3	48.9	1.4
Mask TextSpotter'18 [30]	SynText800k, IC13, IC15, TT	ResNet-50-FPN	52.9	71.8	4.8
Two-stage [36]	SynText800k, IC13, IC15, TT	ResNet-50-SAM	45.0	-	-
TextNet [36]	SynText800k, IC13, IC15, TT	ResNet-50-SAM	54.0	-	2.7
Li et al. [19]	SynText840k, IC13, IC15, TT, MLT, AddF2k	ResNet-101-FPN	57.80	-	1.4
Mask TextSpotter'19 [20]	SynText800k, IC13, IC15, TT, AddF2k	ResNet-50-FPN	65.3	77.4	2.0
Qin et al. [33]	SynText200k, IC15, COCO-Text, TT, MLT Private: 30k (manual label), 1m (partial label)	ResNet-50-MSF	67.8	-	4.8
CharNet [23]	SynText800k, IC15, MLT, TT	ResNet-50-Hourglass57	66.2	-	1.2
TextDragon [43]	SynText800k, IC15, TT	VGG16	48.8	74.8	-
ABCNet-F	SynText150k, COCO-Text, TT, MLT	ResNet-50-FPN	61.9	74.1	22.8
ABCNet			64.2	75.7	17.9
ABCNet-MS			69.5	78.4	6.9

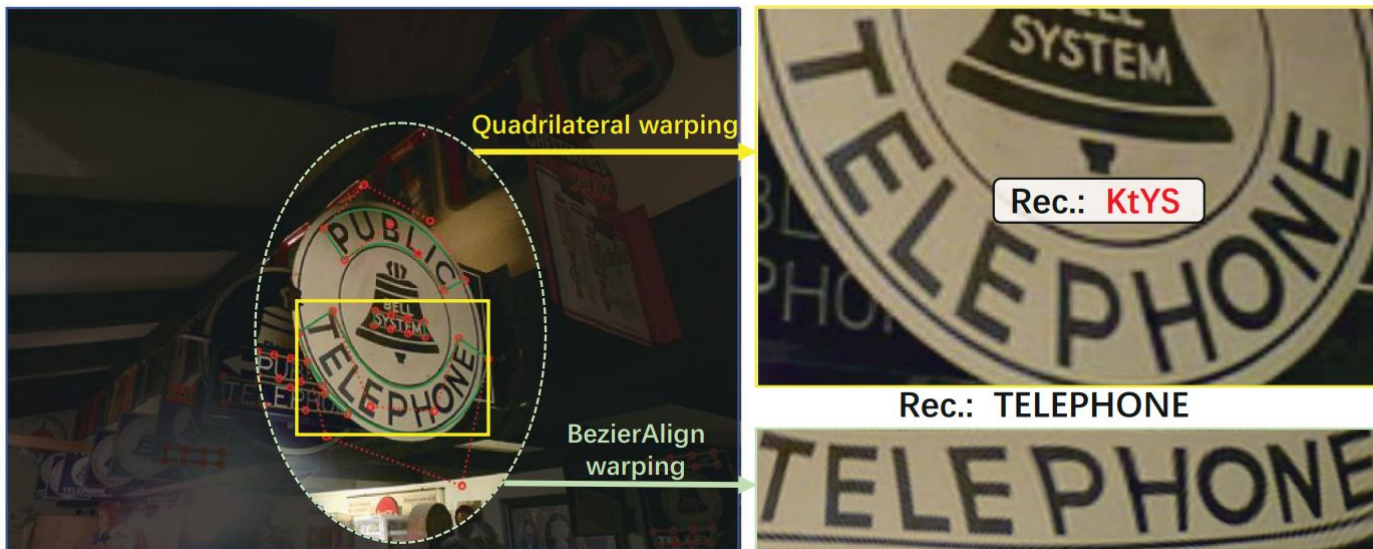
ABCNet

Qualitative results on Total-Text



ABCNet

Bezier Align contribution



ABCNet

Bezier Align contribution



Rec.: PEMEPREE



Rec.: PEACHTREE





ABCNet

Quantitative Bezier Align contribution on Total-Text

Methods	Sampling method	F-measure (%)
ABCNet	with Horizontal Sampling	38.4
ABCNet	with Quadrilateral Sampling	44.7
ABCNet	with BezierAlign	61.9



ABCNet

How expensive is Bezier Align?

Methods	Inference time
without Bezier curve detection	22.8 fps
with Bezier curve detection	22.5 fps



References

- [1] [CHEN, Xiaoxue, et al. Text Recognition in the Wild: A Survey. arXiv preprint arXiv:2005.03492, 2020.](#)
- [2] [LIU, Yuliang, et al. ABCNet: Real-time Scene Text Spotting with Adaptive Bezier-Curve Network. En Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2020. p. 9809-9818.](#)
- [3] [REZATOFIGHI, Hamid, et al. Generalized intersection over union: A metric and a loss for bounding box regression. En Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. 2019. p. 658-666.](#)
- [4] [TIAN, Zhi, et al. Fcos: Fully convolutional one-stage object detection. En Proceedings of the IEEE international conference on computer vision. 2019. p. 9627-9636.](#)
- [5] [GIRSHICK, Ross. Fast r-cnn. En Proceedings of the IEEE international conference on computer vision. 2015. p. 1440-1448.](#)
- [6] [CH'NG, Chee Kheng; CHAN, Chee Seng. Total-text: A comprehensive dataset for scene text detection and recognition. En 2017 14th IAPR International Conference on Document Analysis and Recognition \(ICDAR\). IEEE, 2017. p. 935-942.](#)



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